Tiny Machine Learning for Ultra Low-Power Edge Computing
Summer 2026
Lab 5
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1. In the notebook, the audio waveform is transformed into time-frequency features before being passed to the keyword spotting model. What is the purpose of converting raw audio into spectrogram/MFCC-style features instead of directly using the raw time-domain waveform?

   Converting raw audio into spectrograms/MFCC-style features is it helps the machine better differentiate between different audio samples. Specifically, MFCC uses the mel scale to make it more accurate to how humans hear things rather than just raw data through audio input. Spectrograms just make one sound more distinct from another via turning it to a time-frequency representation, making patterns more prominent.

2. During full integer quantization, the notebook uses a representative dataset. What is the purpose of the representative dataset, and why is it important for creating an INT8 TensorFlow Lite model?

   The representative dataset gives the converter realistic input feature values for integer quantization. When converting a model from 32-bit floating-point values to 8 bit (int 8) the range is reduced dramatically. Having a representative dataset gives us a range (max and min values) that we can make sure get included in that smaller range.

3. What does audio_processor.get_data() do? Refer to input_data.py.

   audio_processor.get_data() acts as a data loader and processor that gets batches of audio features and their ground-truth labels.

4. In the final model export step, the TensorFlow Lite model is converted into a C/C++ byte array. What are g_modeland g_model_len, and why do they need to be copied correctly into the Arduino source file?

   g_model is an array containing the actual raw bytes of the quantized TF lite model, while g_model_len is an integer representing the exact length of the model array.

5. Please include a screenshot showing the correct identification of at least one keyword, either **“YES”** or **“NO”**, as displayed in the Arduino IDE Serial Monitor after deploying your KWS model on the Arduino Nano 33 BLE. The screenshot should be included in your submitted PDF.

```
Heard unknown (203) @2175344ms
Heard unknown (208) @2178288ms
Heard no (207) @2179856ms
Heard unknown (205) @2181056ms
Heard yes (201) @2182624ms
Heard unknown (209) @2183904ms
Heard unknown (225) @2185472ms
```